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ORIGINAL ARTICLE



Postural performance assessment in aging people with diabetes and diabetic peripheral neuropathy using a Wii balance board

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ABSTRACT

Purpose: To determine the effect of sensory perturbations on static postural control in older people with type 2 diabetes mellitus by comparing postural outcomes of people with and without diabetic neuropathy using a Wii Balance Board (WBB).

Materials and methods: Static postural balance assessments were performed in 31 participants: nine with type 2 diabetes mellitus; 12 with diabetic neuropathy; and 10 non-diabetic controls. Participants stood on the WBB under sensory perturbations (visual and proprioceptive). Body balance was analysed using centre of pressure ellipse area, mean velocity, and sample entropy. The effects of within-participant factors, sensory perturbations and the between-participants factor 'group' on outcomes were analysed using a multivariate analysis of variance model.

Results: Type 2 diabetes mellitus participants with and without neuropathy showed altered postural performance under sensory perturbations compared to non-diabetic participants. Moreover, participants with diabetic neuropathy showed impaired postural performance when one perceptual system was disturbed. Finally, participants with type 2 diabetes mellitus without neuropathy decreased their postural performance when both sensory disturbances were present.

Conclusions: The Wii Balance Board can be a useful alternative for balance impairment screening related to diabetic neuropathy and contribute as an affordable source of insight in early interventions in integral diabetes care.

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KEYWORDS

Diabetic neuropathy; postural balance; diabetes mellitus; aged; diabetes complications

► IMPLICATIONS TO REHABILITATION

- Older people with diabetic peripheral neuropathy depend on visual and somatosensory cues to keep their static postural balance.
- Static balance assessment using the Wii Balance Board allows the identification of alterations in postural performance in participants with diabetes.
- This low-cost method used can be considered as a complement to integral diabetes care.

Introduction

Diabetes constitutes one of the main health conditions of world-wide concern [1]. According to global health estimates in 2019–2020 from the World Health Organization, diabetes mellitus is one of the main causes of death worldwide [2]. Furthermore, diabetic neuropathy (DN) is the most prevalent complication associated with type 2 diabetes mellitus (T2DM), affecting up to 50% of individuals with diabetes [3]. DN is characterised by damage to the neural tissue of peripheral nerves, retina and vestibular system, compromising proprioception from the lower limbs, as well as spatial orientation and balance [4]. Thus, T2DM and DN related impairment can increase fall risk and lead to decreased quality of life, disability and institutionalisation in older people [5–7].

As shown in previous studies, it is possible to assess some pathologies by means of the alterations caused to the patient's coordination capacities, especially in postural control. Previous research on postural control impairments related to T2DM in older adults (people aged 60 and older) showed decline in

reactive balance control associated with muscle weakness and plantar insensitivity related to neuropathy [8]. Moreover, T2DM has been linked to subclinical imbalance disturbances before the manifestation of any clinical signs of sensory degradation [9], particularly in challenging static body balance conditions [10,11]. When DN was present, patients exhibited significant deficits in sensorimotor function, balance and gait [12], and impaired postural control in bipedal stance when sensory cues were perturbed [13]. Additionally, impaired postural parameters have been shown to be associated with altered nerve velocity conduction in adults with DN [14].

Given this context, instrumented balance assessments can be a useful tool for the early diagnosis of balance impairments in older people with DN in daily clinical care. However, due to the complexity and high cost of instruments required for gold-standard studies (i.e., force platforms), quantitative postural assessments are limited to research facilities and their use in health services is precluded [7,15]. As a low-cost alternative device to force platforms, the Nintendo Wii Balance Board (WBB) provide comparable

data to assess body sway through the analysis of centre of pressure (COP) path length during static balance trials [15]. The WBB has been shown to be sensitive in detecting COP behaviour in visual tasks during standing in non-diabetic older adults [16], and was proposed as a valid and reliable method for quantifying postural performance in older adults with T2DM [17].

To explore whether expected postural control deficits in older people with diabetes and related complications (i.e., with or without DN) can be identified using a WBB, the aim of this study was to determine the effect of visual and proprioceptive perturbations in static postural control in older people with diabetes with or without diabetic neuropathy, and in aging non-diabetics.

Materials and methods

Study design, setting and participants

This cross-sectional study was conducted at the Human Movement Analysis Laboratory of the Universidad de Antofagasta. This investigation was designed following the Strengthening the Reporting of OBservational studies in Epidemiology (STROBE) statement [18]. All participants signed an informed consent form before their enrolment in the research protocol. The research protocol was approved by the Scientific Research Ethics Committee of Antofagasta University (CEIC-UA, n°155/2018) and all requirements set out in the Declaration of Helsinki (1975, subsequently revised in 2008) were also met.

Non-probabilistic convenience sampling was used for participant enrolment. Thirty-one participants aged 55 and older were recruited for this investigation, 10 of whom were non-diabetic and 21 of whom had type 2 diabetes mellitus. Participants were assigned to one of the following groups: control group (CG; $n = 10$; 6 males/4 females), T2DM group (T2DMG $n = 9$; 5 females/4 males) and T2DM with DN group (DNG $n = 12$; 4 females/8 males).

The inclusion criteria for T2DMG were: >2 years since T2DM diagnosis by a physician and independence in carrying out life activities according to the Barthel Index Score [19]. Furthermore, all participants were screened by the same evaluator for normal blood pressure levels and hypoglycaemia using a glucometer (Abbot, Freestyle-neo, FSL, Abbott Diabetes Care, Alameda, CA), and peripheral neuropathy before postural assessment. Participants were excluded if any of the following exclusion criteria were met: any neurological condition other than diabetic neuropathy, vestibular disorders, uncontrolled hypertension, cognitive impairment according to the Minimental test [20], vision loss, hypoglycaemia, amputation, or any traumatic lower limb injuries in the past 6 months.

Participants with T2DM were assigned to DNG if two of the following tests were positive (abnormal):

1. Diabetic Neuropathy Symptom score of 1 or more [21].
2. Examination with 10g Semmes-Weinstein monofilament for touch sensation over the 1st, 3rd and 5th metatarsal heads and plantar surface of the distal hallux.
3. Vibration perception with a 128Hz tuning fork over the dorsum of the first toe proximal to the nail bed.

Monofilament and vibration perception tests were assessed in both feet according to Yang et al. [22].

Procedure

To measure static postural performance, a Nintendo Wii Balance Board (WBB; Nintendo, Kyoto, Japan) device connected by

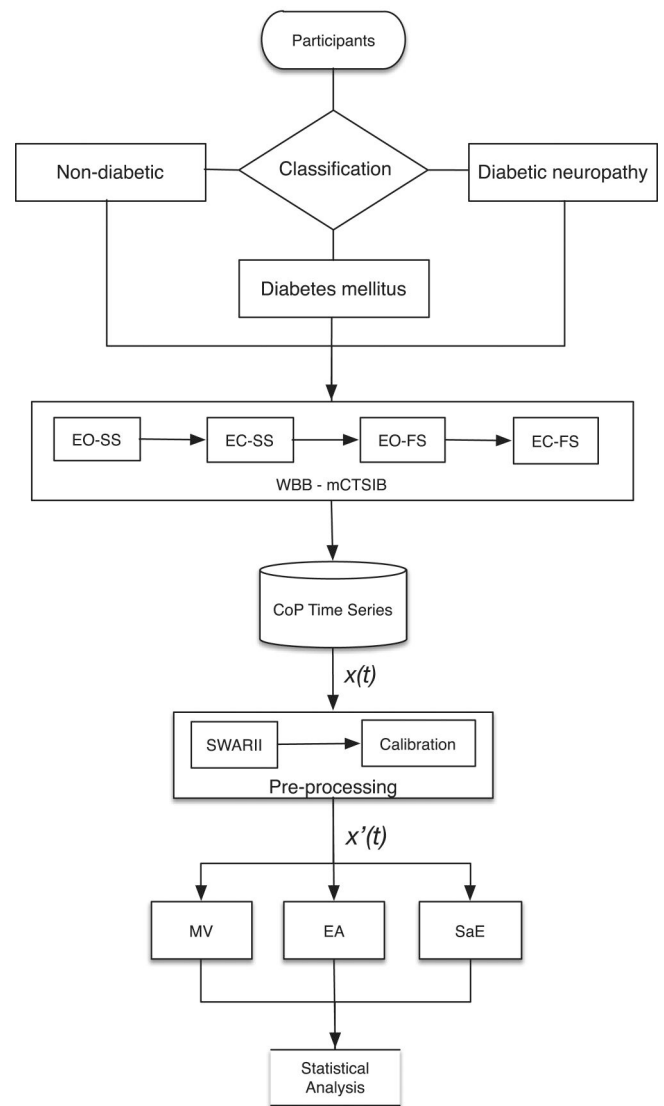


Figure 1. Protocol flow chart. EO: eyes open; EC: eyes closed; SS: stable surface; FS: foam surface; WBB: Wii Balance Board; mCTSIB: modified Clinical Test of Sensory Interaction for Balance; CoP: centre of pressure; SWARII: sliding window average with relevance interval interpolation; MV: mean velocity; EA: ellipse area; SaE: sample entropy.

Bluetooth to a laptop was used to record COP displacement in medial-lateral and anterior-posterior sway directions. COP signals were acquired at a sampling rate of 100 Hz. Data was extracted using the interface software BrainBLoX [23].

Participants stood on the WBB during the assessment. Four increasingly challenging balance conditions were applied, according to an instrumented version of the modified Clinical Test of Sensory Interaction for Balance [24]. The four conditions were bipedal stance with: Eyes open (EO) on stable surface (SS); Eyes closed (EC) on SS; EO on a low-density foam surface (FS); and EC on FS. Participants were instructed to stand barefoot and place both feet over reference marks on the WBB, to keep their hands alongside their body and to stand as still as possible while looking straight ahead at a reference mark placed on the wall. See Figure 1 for a flow chart summarising the study protocol.

Participants performed three 30-s trials in each condition, with 20-s of sitting rest between them. Since this kind of procedures does not show a significant learning effect in older people with

balance disorders [25,26], the protocol of this research was applied in a single session.

Signal processing

To address inconsistent sampling rates reported from WBB signals [15], COP time series were resampled using a sliding window average, with relevance interval interpolation implemented in Python [27]. To reduce COP measurement error and provide more reliable estimates of time-domain COP features, signals were then imported into Matlab R2017b (The MathWorks Inc. Natick, Massachusetts), where linear calibration of each COP signal was applied following the recommendations of Leach et al. (2014). For this purpose, an m -coefficient (i.e., slope of the trend lines in the linear equation) was defined as a factor for each value in each COP time series. For values on the medial-lateral axis, an m value of 1/1.098 was applied, while for values on the anterior-posterior axis, an m -value of 1/1.088 was selected, since they showed lower measurement error when compared to a force platform [28].

Body balance was analysed in terms of three COP measures: 95% confidence ellipse area (EA) of COP displacement; mean COP velocity in the mediolateral (MVx) and anteroposterior (MVy) axis; and sample entropy in mediolateral (SaEx) and anteroposterior (SaEy) COP displacement. EA is considered representative of overall postural performance [29], while mean COP velocity characterises the net muscular activity of the postural control system and has been associated with its efficiency [30]. As a nonlinear parameter, sample entropy quantifies the temporal structure of a signal and represents the regularity of the COP time series. In balance assessment conditions, increased postural regularity has been associated with aging, disease or injuries, and less adaptive capabilities of the postural system [31]. The input parameters used in sample entropy calculations were $m = 2$ and $r = 0.15 \times$ standard deviation [32].

Data analysis

Sample size calculation was performed in G*Power version 3.1.9.4. (G*Power, Version 3.0.10, Franz Faul, Universität Kiel, Kiel, Germany) [33], following previous recommendations [34]. Considering a Pillai-Bartlett trace value of 0.4, a significance level $\alpha = 0.05$ and a power $\beta = 0.8$, a sample of 27 participants was required. Further statistical analysis was performed using SPSS v.25 (IBM Corp., Armonk, New York).

The Shapiro-Wilk test was used to determine data distribution and Levene's test was applied to determine homogeneity of variance. To determine the effects of within-participant factors, the 'visual' (EO and EC) and 'proprioceptive' (SS and FS) systems, and that of the between-participant factor 'group' (CG, DNG and T2DMG) on COP outcomes (EA, MVx, MVy, SaEx and SaEy), a multivariate analysis of variance mixed model was applied. The follow-up was carried out by means of univariate comparisons.

Finally, in the case of significant effects, pairwise comparisons with Bonferroni correction were applied. For all statistical tests, an alpha of 0.05 was used.

Results

Baseline demographics and characteristics of the participants are reported in Table 1. Significant differences between groups were observed only in glucose levels.

Regarding the main study analyses, multivariate comparisons revealed a main effect of group ($F_{10,88} = 4.02$; $p < 0.001$; $\eta^2_p = 0.31$), visual constraints ($F_{2,43} = 6.68$; $p < 0.001$; $\eta^2_p = 0.44$) and proprioceptive constraints ($F_{5,43} = 17.21$; $p < 0.001$; $\eta^2_p = 0.67$) on postural control variables. Moreover, two interaction effects were found on the dependent variables: group $\times$ visual constraints ($F_{10,88} = 2.14$; $p = 0.03$; $\eta^2_p = 0.2$) and visual constraints $\times$ proprioceptive constraints ($F_{5,43} = 4.51$; $p = 0.002$; $\eta^2_p = 0.34$). The rest of interaction effects were not statistically significant. The univariate comparisons related to these effects are shown in Table 2. As can be seen, the effects on EA, MVy and MVx were significant, but not on sample entropy (x- nor y-axis). Only a main effect of group was found in SaEy.

Table 2. Univariate effects on the postural control variables.

Effect	Variable	Degrees of freedom	F-value	p Value	η^2_p
Group	EA (cm ²)	2; 47	12.02	<0.001	0.34
	MVx (cm/s)	2; 47	12.52	<0.001	0.35
	MVy (cm/s)	2; 47	10.17	<0.001	0.3
	SaEx	2; 47	1.23	0.3	0.05
	SaEy	2; 47	6.87	0.002	0.23
Visual constraints	EA (cm ²)	1; 47	14.09	<0.001	0.23
	MVx (cm/s)	1; 47	11.49	0.001	0.2
	MVy (cm/s)	1; 47	11.86	0.001	0.2
	SaEx	1; 47	0.88	0.35	0.02
	SaEy	1; 47	24.83	<0.001	0.35
Proprioceptive constraints	EA (cm ²)	1; 47	64.34	<0.001	0.58
	MVx (cm/s)	1; 47	67.27	<0.001	0.59
	MVy (cm/s)	1; 47	27.89	<0.001	0.37
	SaEx	1; 47	1.11	0.3	0.02
	SaEy	1; 47	1.45	0.23	0.03
Visual and proprioceptive constraints	EA (cm ²)	1;47	20.38	<0.001	0.3
	MVx (cm/s)	1;47	8.83	0.005	0.16
	MVy (cm/s)	1;47	3.08	0.09	0.06
	SaEx	1;47	0.11	0.74	0.002
	SaEy	1;47	0.01	0.91	<0.001
Group $\times$ visual constraints	EA (cm ²)	2; 47	3.85	0.03	0.14
	MVx (cm/s)	2; 47	3.93	0.03	0.14
	MVy (cm/s)	2; 47	3.35	0.04	0.12
	SaEx	2; 47	1.9	0.16	0.07
	SaEy	2; 47	0.48	0.62	0.02

EA: ellipse area; MVy: mean COP velocity in anterior-posterior direction; MVx: mean COP velocity in medial-lateral direction; SaEy: sample entropy anterior-posterior COP displacement; SaEx: sample entropy medial-lateral COP displacement; η^2_p : Partial eta square.

Table 1. Demographic characteristics.

Variable	All (n = 31)	CG (n = 10)	T2DMG (n = 9)	DNG (n = 12)	p Value
Age, years	63.69 $\pm$ 8.99	60.27 $\pm$ 8.60	65.67 $\pm$ 11.10	66.13 $\pm$ 6.79	0.213
Weight, kg	73.51 $\pm$ 12.52	73.77 $\pm$ 10.60	79.61 $\pm$ 7.84	68.44 $\pm$ 18.57	0.936
Height, m	1.62 $\pm$ 0.08	1.62 $\pm$ 0.07	1.62 $\pm$ 0.08	1.61 $\pm$ 0.09	0.179
BMI, kg/m ²	28.06 $\pm$ 4.29	28.03 $\pm$ 3.88	30.32 $\pm$ 2.59	26.24 $\pm$ 6.31	0.281
DM diagnosis, years	12.29 $\pm$ 13.78	–	21.89 $\pm$ 11.06	18.38 $\pm$ 13.94	0.571
Glucose, mg/dl	146.36 $\pm$ 68.85	99.09 $\pm$ 13.44	149.78 $\pm$ 55.33	207.50 $\pm$ 82.01	<0.001

Data expressed as mean $\pm$ standard deviation. BMI: body mass index; DM: diabetes mellitus; CG: control group; T2DMG: type 2 diabetes mellitus group; DNG: diabetic neuropathy group. For the DM diagnosis variable, an independent t-test was used for T2DMG and DNG comparison.

Statistically significant differences between groups are shown in Figures 2 and 3. Overall, participants with diabetic neuropathy showed increased COP linear variables (EA, MVx and MVy) compared to diabetic and non-diabetic participants when vision was challenged in all surface conditions. Furthermore, diabetic participants exhibited impaired postural performance (i.e., increased COP variable values) compared to non-diabetic participants only when visual and proprioceptive information were simultaneously challenged. (i.e., EC/FS) (Figures 2–3).

Non-diabetic participants did not show significant changes in COP linear outcomes between visual feedback conditions, while participants with diabetic neuropathy showed significant differences between the open and closed eyes conditions, both on the stable and foam surface. Overall, the interaction between group and visual constraints showed that the restriction of visual information decreased postural control in the diabetic neuropathy and

type 2 diabetes mellitus groups (only in the foam condition), but not in non-diabetic participants.

Regarding sample entropy, a higher SaEy was observed in the two groups of participants with diabetes when compared to non-diabetics ($p < 0.05$), regardless of visual and proprioceptive constraints.

Discussion

The aim of this research was to determine the effect of sensory perturbations on static postural control in older people with diabetes with or without diabetic neuropathy, and in aging non-diabetics using data from the WBB. Our results suggest that older adults with diabetes exhibit poorer postural performance only when perceptual systems are disrupted. Moreover, regarding the effect of sensory disturbance, disruption of one system (visual or proprioceptive) has a negative effect on postural performance in people with diabetic neuropathy (i.e., increased EA, MVx and MVy), while people with type 2 diabetes showed postural instability when both sensory disturbances were present (i.e., eyes closed over foam surface).

Results obtained with the WBB were consistent with other studies based on force platform data, where participants with diabetic neuropathy showed poorer postural control when compared with non-diabetic aging participants [35–37]. The increased COP velocity (MVx, MVy) and displacement area (EA) may reflect sensory impairment in people with diabetic neuropathy [38] and may be related to visual dependence and lack of sensory integration. Similar results were reported by Shin et al. (2021), where increased COP distance in both axes with the eyes closed was strongly associated with abnormal sensory nerve conduction in the sural nerve, while mediolateral COP distance during both eye conditions was correlated with muscle action potential amplitude in the peroneal nerve in people with diabetic neuropathy [14]. Moreover, Morrison et al. (2012) reported differences in COP variables between people with and without diabetes with fall risk when standing over a foam surface with eyes closed, suggesting that fallers with diabetes present greater postural sway velocity (i.e., MVy in our research) [5]. Finally, regarding sample entropy, an absence of changes in regularity were observed in non-diabetic participants, which is consistent with previous research

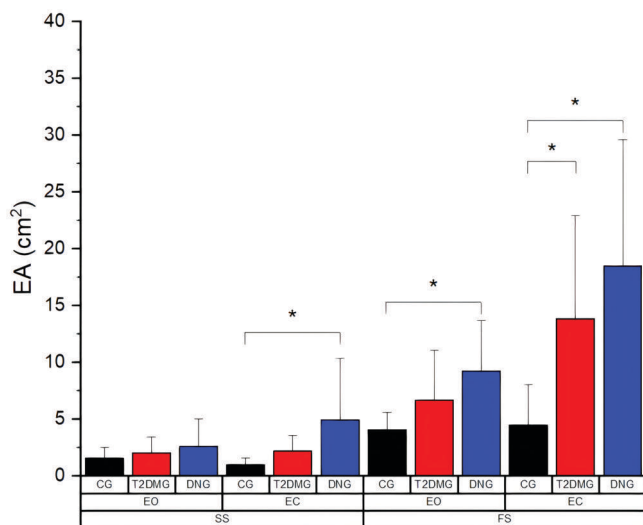


Figure 2. Centre of pressure ellipse area (EA) results between groups and sensory conditions. The bars represent the median and error bars represent the standard deviation. *Indicates significant differences ($p < 0.05$). CG: control group; T2DMG: type 2 diabetes mellitus group; DNG: diabetic neuropathy group; EO: eyes open; EC: eyes closed; SS: stable surface; FS: foam surface.

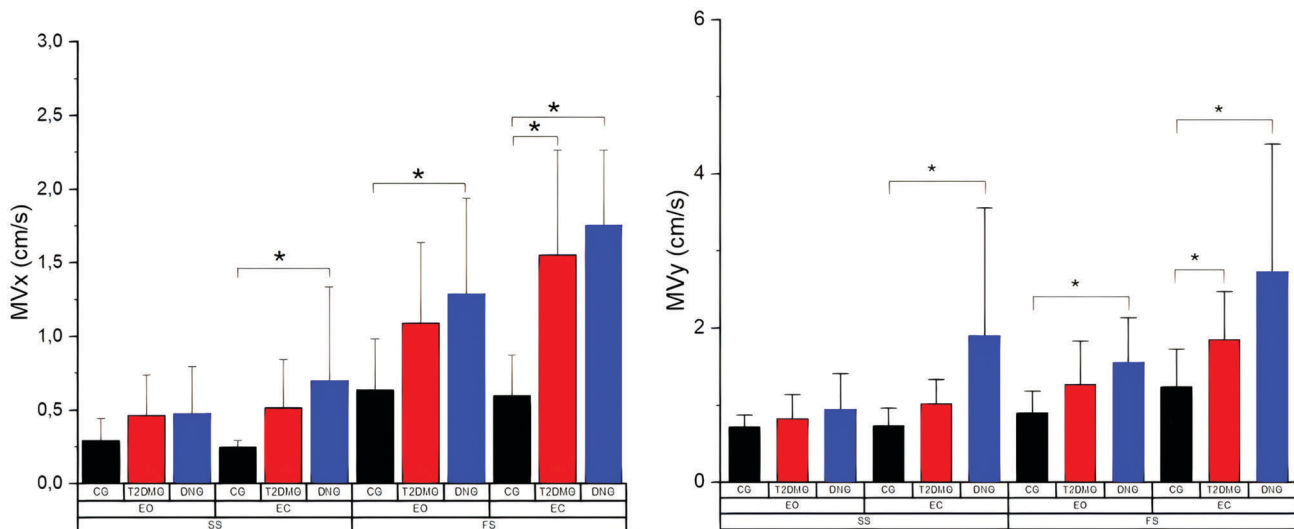


Figure 3. Centre of pressure mean velocity results between groups and sensory conditions in the: A) medial-lateral axis (MVx) and B) anterior-posterior axis (MVy). *Indicates significant differences ($p < 0.05$). Bars represent the mean and error bars represent the standard deviation. CG: control group; T2DMG: type 2 diabetes mellitus group; DNG: diabetic neuropathy group; EO: eyes open; EC: eyes closed; SS: stable surface; FS: foam surface.

[39,40]. Moreover, decrease in body sway displacement signal regularity in the anterior-posterior axis (i.e., higher COP SaEy) has been previously reported in older people with both type 2 diabetes mellitus and high fall risk [5].

As postural deficit screening is an integral component of clinical assessment of the elderly, the proposed method quantifies changes in postural control in older people, and could be helpful for selecting therapeutic strategies that have proven to be useful in the management of diabetes and diabetic neuropathy [41] and for the evaluation of the impact of such interventions according to the characteristics of the individual.

The present study has some limitations that should be considered. Firstly, our research was a cross-sectional observational study with a low sample size. Secondly, the DN assessment method did not allow a quantitative evaluation of sensory small diameter afferent fibre function and other factors involved in balance disorders in older people, such as the presence of sarcopenia. Future studies should consider the use of screening tools to assess functional compromise of small and large diameter sensory fibres and loss of skeletal muscle mass to determine their contribution in alterations of postural balance performance. Finally, the use of the WBB also involves some technical limitations as it is not a device designed for clinical purposes; however, the recommended processing methods were used to mitigate these constraints.

Conclusions

While non-diabetic older people do not experience changes in static postural control in conditions with sensory restriction, the presence of type 2 diabetes mellitus compromises balance performance in challenging sensory conditions. Participants with diabetic peripheral neuropathy showed significant postural balance impairment when slight sensory constraints were present. Furthermore, a WBB-based method can be a low-cost alternative for screening of static balance performance impairments in diabetic patients and contribute as an affordable insight into early exercise-based interventions in diabetes care.

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Disclosure statement

The authors have no conflicts of interest to declare.

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